TAM 470 Project 2 Report

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Project problem description:

Governing PDE: The 2D transient heat equation with constant material properties and a source term is:

$$\rho c \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + Q(x, y, t) \tag{1}$$

In Equation (1), ρ is mass density, c is heat capacity, k is thermal conductivity, T is the temperature and Q is the heat source term. The PDE holds for a rectangular domain:

$$x \in (0, L_x), \quad y \in (0, L_y) \tag{2}$$

Initial conditions: At time 0, the entire domain has a uniform temperature $\bar{T}$:

$$T(x, y, 0) = \bar{T} \tag{3}$$

Boundary conditions: The bottom of the domain is fixed at the initial temperature $\bar{T}$:

$$T(x, 0, t) = \bar{T} \tag{4}$$

The left, right, and top sides of the domain lose heat to the surroundings, and the heat loss is dominated by radiation heat flux:

$$k \frac{\partial T}{\partial n}(0, y, t) = q_{rad} \tag{5a}$$

$$k \frac{\partial T}{\partial n}(L_x, y, t) = q_{rad} \tag{5b}$$

$$k \frac{\partial T}{\partial n}(x, L_y, t) = q_{rad} \tag{5c}$$

In Equations (5), k is thermal conductivity and $\frac{\partial T}{\partial n}$ is the temperature gradient in the normal direction on the surface. The radiation heat flux is defined as:

$$q_{rad} = -\sigma \epsilon (T^4 - T_\infty^4) \tag{6}$$

In Equation (6), σ is the Steffan-Boltzmann constant, ϵ is the emissivity factor, and T_∞ is the ambient temperature.

Problem schematic: A schematic of the domain and laser, which scans the top of the domain and moves with a constant velocity v_L, is shown in Figure 1.

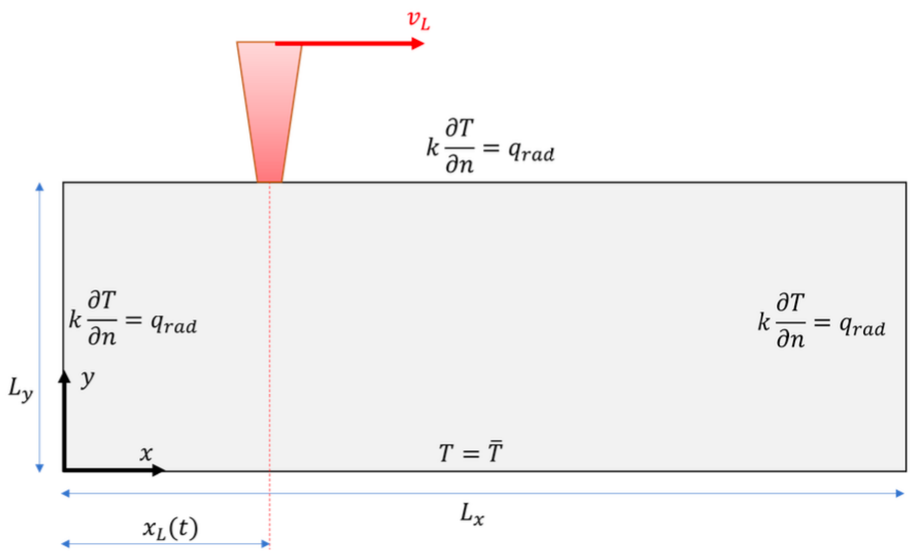

Figure 1: Domain and boundary conditions for 2D domain.

Laser heating model: A common representation of laser heating is that of a heat source term with a Gaussian intensity profile that decays away from the center of the laser, and an exponential decay profile with depth into the surface of the material being heated. Therefore, the heat source term Q is represented by the following function:

$$Q(x,y,t) = \left(\frac{P}{w^2\sqrt{2\pi}}\right)\exp\left(\frac{-r^2}{2w^2}\right)f(y) \tag{7}$$

In Equation (7), P represents the laser power, w is a measure of the width of the beam called the $\frac{1}{e^2}$ *spot size*, and r is the distance away from the laser center x_L.

Note: a fully 3D laser heating model has a 2D Gaussian distribution at the surface. For our 2D simplified treatment of the problem, the Gaussian is 1D and we are essentially assuming the rectangular domain has thickness w into the page (z direction). The laser's heat is therefore distributed with a Gaussian profile in the x direction and a uniform profile in the third z direction. This is simplification is necessary for our 2D model of the problem.

The laser's focus is assumed at any time to be situated at the coordinates $(x_L(t), L_y)$, such that we can write

$$r^2 = (x - x_L(t))^2 \tag{8}$$

The function $f(y)$ dictates the rate of decay of laser heat penetration into the material. For this model, we assume:

$$f(y) = \frac{\beta}{L_y}\exp\left(-\beta\left(\frac{d}{L_y}\right)\right), \quad d = L_y - y \tag{9}$$

where β is a penetration decay parameter.

Laser motion: You can assume the laser center begins at position x_{start} and moves at a constant velocity v_L in the $+x$ direction when $v_L > 0$.

Fixed simulation parameters: Table 1 lists parameters that you can assume are **fixed for all simulations** that you conduct in this project.

Parameter	Description	Value	Units
L_x	Domain length in x direction	.05	m
L_y	Domain length in y direction	.03	m
w	Laser spot size	0.004	m
$\bar{T}$	Initial and fixed temperature	298	K
T_∞	Ambient temperature	298	K
k	Thermal conductivity	48	$\mathrm{W/(m \cdot K)}$
ρ	Mass density	7900	$\mathrm{kg/m^3}$
c	Specific heat capacity	470	$\mathrm{W/(kg \cdot K)}$
σ	Steffan-Boltzmann constant	5.67×10^{-8}	$\mathrm{W/(m^2 \cdot K^4)}$
ϵ	Emissivity factor	0.8	Dimensionless
β	Penetration decay parameter, see Equation (9)	2	Dimensionless

Table 1: Table of fixed parameters for all simulations. Material properties are approximately those associated with steel.

Question 1:

Question 1 (20 points):

Write down the numerical scheme required to solve the transient heat Equation (1) on the two-dimensional domain using central differences for discretizing the PDE, first-order difference approximations for flux boundary conditions, and the forward Euler scheme for time integration. Your scheme should assume uniform (but not necessarily equal) grid spacings Δx and Δy (with $N_x + 1$ and $N_y + 1$ grid points in the x and y dimensions, respectively) in space and constant time steps of size Δt for time integration. Specifically, be sure to describe:

1. Your update scheme for interior nodes.
2. Your update scheme for boundary nodes, including details on how you will handle the non-linear radiation boundary condition in Equation (5)–(6).

Part1: 2D transient heat equation with constant material properties and a source term is:

$$\rho c * \frac{\partial T}{\partial t} = k * \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + Q(x, y, t) \tag{1}$$

$$\frac{\partial T}{\partial t} = \frac{k}{\rho c} * \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + \frac{Q(x,y,t)}{\rho c} \tag{2}$$

By applying central differences for spatial derivative

$$\frac{\partial^2 T}{\partial x^2} = \frac{T^n{}_{i+1,j} - 2T^n{}_{i,j} - T^n{}_{i-1,j}}{\Delta x^2} \tag{3}$$

$$\frac{\partial^2 T}{\partial y^2} = \frac{T^n{}_{i,j+1} - 2T^n{}_{i,j} - T^n{}_{i,j-1}}{\Delta y^2} \tag{4}$$

Plug (4) back to (2) and use forward-euler scheme can get the update scheme for interior nodes:

$$T^{n+1} = \frac{k * \Delta t}{\rho c} * \left(\frac{T^n{}_{i+1,j} - 2T^n{}_{i,j} - T^n{}_{i-1,j}}{\Delta x^2} + \frac{T^n{}_{i,j+1} - 2T^n{}_{i,j} - T^n{}_{i,j-1}}{\Delta y^2}\right) + \frac{\Delta t * Q(x,y,t)}{\rho c} + T^n \tag{5}$$

Part 2: 4 boundary conditions and qrad are:

$$\frac{\partial T}{\partial t}(0,y,t) = \frac{qrad}{k}\ (Left\ side) \tag{6}$$

$$\frac{\partial T}{\partial t}(Lx,y,t) = \frac{qrad}{k}\ (Right\ side) \tag{7}$$

$$\frac{\partial T}{\partial t}(x,Ly,t) = \frac{qrad}{k}\ (Top\ side) \tag{8}$$

$$T(x,0,t) = \tilde{T}\ (Bottom\ side) \tag{9}$$

$$qrad = -\sigma\epsilon\left(T^4 - T_{inf}^4\right) \tag{10}$$

The bottom side boundary condition is a diritchlet condition which do not need to modify overtime and will always follow equation (9).
For left,right, and top side, the boundary condition are Neumann condition, which need to be modified over time, and below shows how to handle these boundary conditions. For instance, for the Left side:

$$\frac{T_{0,j}^n - T_{1,j}^n}{\Delta x} = \frac{qrad}{k} = \frac{-\sigma\epsilon\left(T^4 - T_{inf}^4\right)}{k} \tag{11}$$

$$\frac{T_{0,j}^n - T_{1,j}^n}{\Delta x} + \frac{\sigma\epsilon\left(T^4 - T_{inf}^4\right)}{k} = 0 \tag{12}$$

In equation (12), n represent the time indicator, and equation (12) is a non-linear ODE becaue of power of 4. In order to solve non-linear ODE, Newton-Raphson method is used.

Newton-Raphson method states that:

$$T_{0,j,n+1}^{k+1} = T_{0,j,n+1}^{k} - \frac{f\left(T_{0,j,n+1}^{k}\right)}{f'\left(T_{0,j,n+1}^{k}\right)} \tag{13}$$

Which left hand side of equation (12) is the $f\left(T_{0,j,n+1}^{k}\right)$:

$$f\left(T_{0,j,n+1}^{k}\right) = \frac{T_{0,j,n+1}^{k} - T_{1,j,n+1}^{k}}{\Delta x} + \frac{\sigma\epsilon\left({T_{0,j,n+1}^{k}}^4 - T_{inf}^4\right)}{k} \tag{14}$$

And therefore $f'\left(T_{0,j,n+1}^{k}\right)$ is:

$$f'\left(T_{0,j,n+1}^{k}\right) = \frac{1}{\Delta x} + \frac{\sigma\epsilon\left(4{T_{0,j,n+1}^{k}}^3\right)}{k} \tag{15}$$

Follow the same ideaology, the equations to handle top and right side boundary conditions can be obtained.

For right side:

$$T_{Nx,j,n+1}^{k+1} = T_{NX,j,n+1}^{k} - \frac{f\left(T_{Nx,j,n+1}^{k}\right)}{f'\left(T_{Nx,j,n+1}^{k}\right)} \tag{16}$$

$$f\left(T_{Nx,j,n+1}^{k}\right) = \frac{T_{Nx,j,n+1}^{k} - T_{Nx-1,j,n+1}^{k}}{\Delta x} + \frac{\sigma\epsilon\left({T_{Nx,j,n+1}^{k}}^4 - T_{inf}^4\right)}{k} \tag{17}$$

$$f'\left(T_{Nx,j,n+1}^{k}\right) = \frac{1}{\Delta x} + \frac{\sigma\epsilon\left(4{T_{Nx,j,n+1}^{k}}^3\right)}{k} \tag{18}$$

For top side:

$$T_{i,Ny,n+1}^{k+1} = T_{i,Ny,n+1}^{k} - \frac{f\left(T_{i,Ny,n+1}^{k}\right)}{f'\left(T_{i,Ny,n+1}^{k}\right)} \tag{19}$$

$$f\left(T_{i,Ny,n+1}^{k}\right) = \frac{T_{i,Ny,n+1}^{k} - T_{i,Ny-1,n+1}^{k}}{\Delta y} + \frac{\sigma\epsilon\left({T_{i,Ny,n+1}^{k}}^{4} - T_{inf}^{4}\right)}{k} \tag{20}$$

$$f'\left(T_{i,Ny,n+1}^{k}\right) = \frac{1}{\Delta x} + \frac{\sigma\epsilon\left(4{T_{i,Ny,n+1}^{k}}^{3}\right)}{k} \tag{21}$$

Question 2:

Question 2 (20 points):

Write a code to model a stationary laser that inputs heat at the midpoint of the top of the domain: $x = 0.5L_x$, $y = L_y$. Consult Table 2 below for additional parameters to use in your simulation.

Parameter	Description	Value	Units
P	Laser power	800	Watts
N_x	$N_x + 1$ grid points in the x direction	50	None
N_y	$N_y + 1$ grid points in the y direction	50	None
t_{end}	Simulation end time	150	sec

Table 2: Simulation parameters for Question 2.

Choose a time step that gives stable and converged results in time (you can assume the results in space are sufficiently converged). Give some justification or results to defend your choice of time step.

Submit the following

1. (4 pts) A table listing the maximum temperature at the top surface of the domain at $t = 30, 60, 90, 120, 150$ seconds.
2. (4 pts) Plots of temperature vs time at $x = 0.5L_x$ at the top surface ($y = L_y$), 20% depth ($y = 0.8L_y$), and 50% depth ($y = 0.5L_y$). You may plot all three curves on the same figure if you wish.
3. (4 pts) A plot of temperature vs position, at the final time, along the line $x = 0.5L_x$: that is, plot $T(0.5L_x, y, t_{end})$ vs y.
4. (4 pts) Contour plots of temperature distribution throughout the domain at $t = 10, 40, 150$ seconds (3 total plots).
5. (4 pts) Write a few sentences summarizing the key aspects of your simulation results and whether they make physical sense.

Part 1.

Time (s)	Temperature (K)
30	1661.39
60	1880.54
90	1932.00
120	1943.73
150	1946.39

Table 1. Maximum temperature at top surface at different time

Part 2.

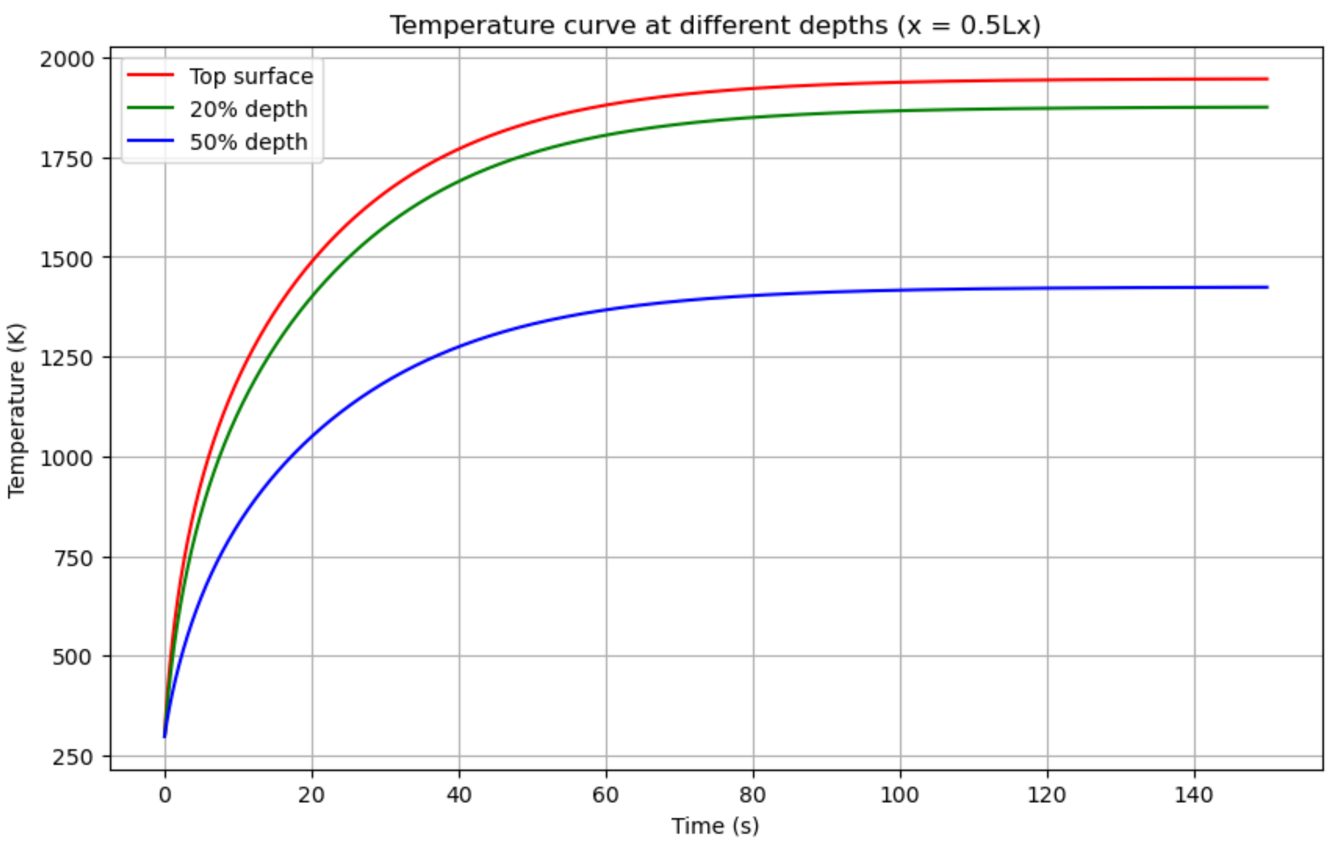

Figure 1. Plot of temperature vs time at x=0.5Lx at top with different depth

Part 3.

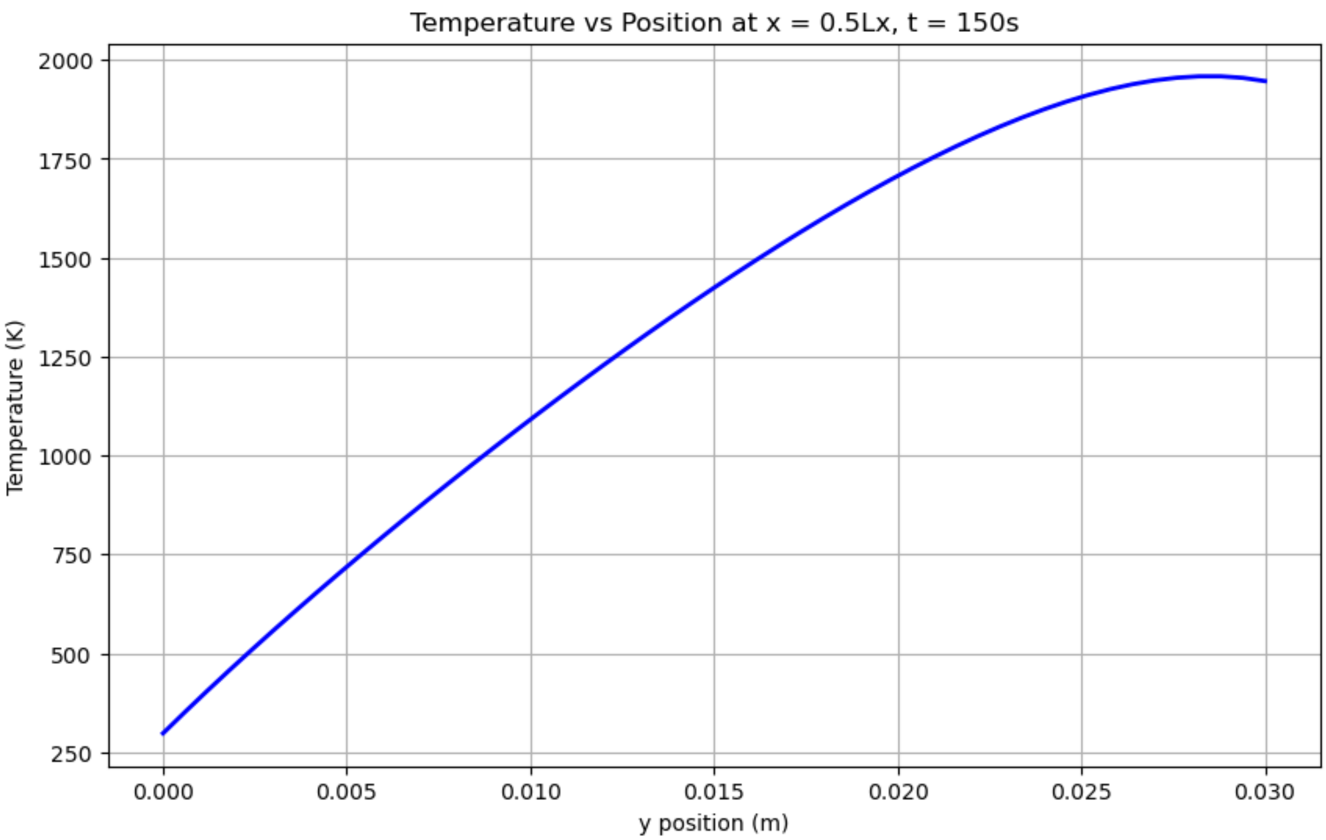

Figure 2. Temperature vs position, at the final time, along x=0.5Lx

Part 4.

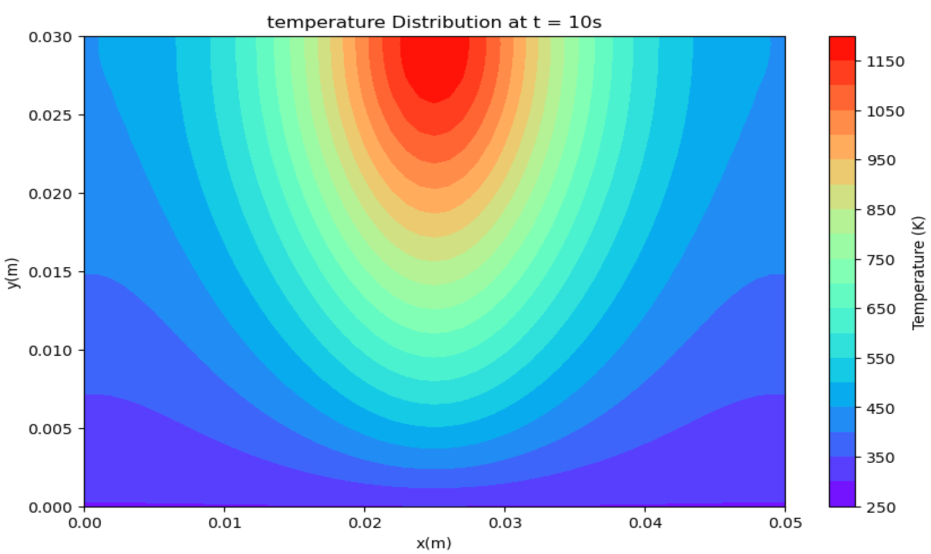

Figure 3. Temperature contour plot at t=10s

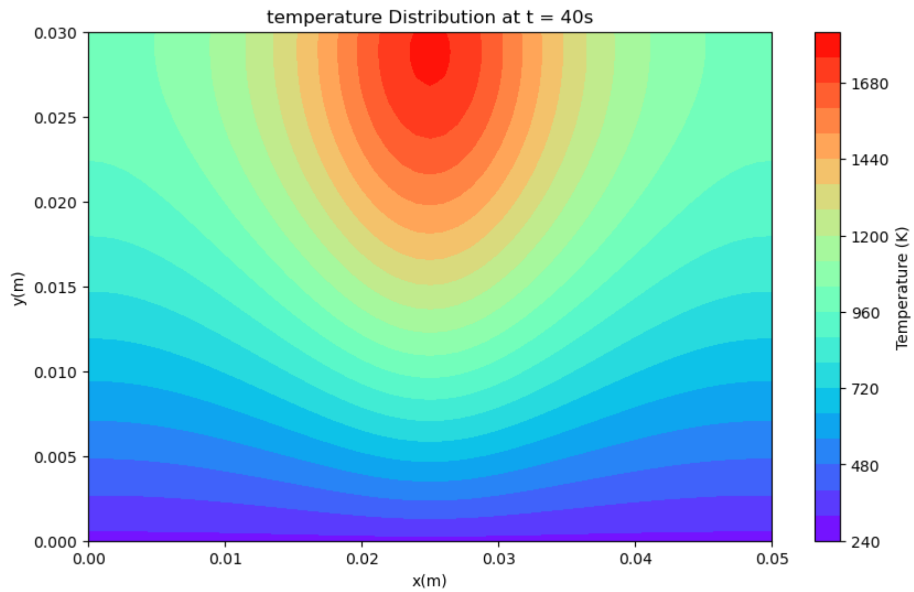

Figure 4. Temperature contour plot at t=40s

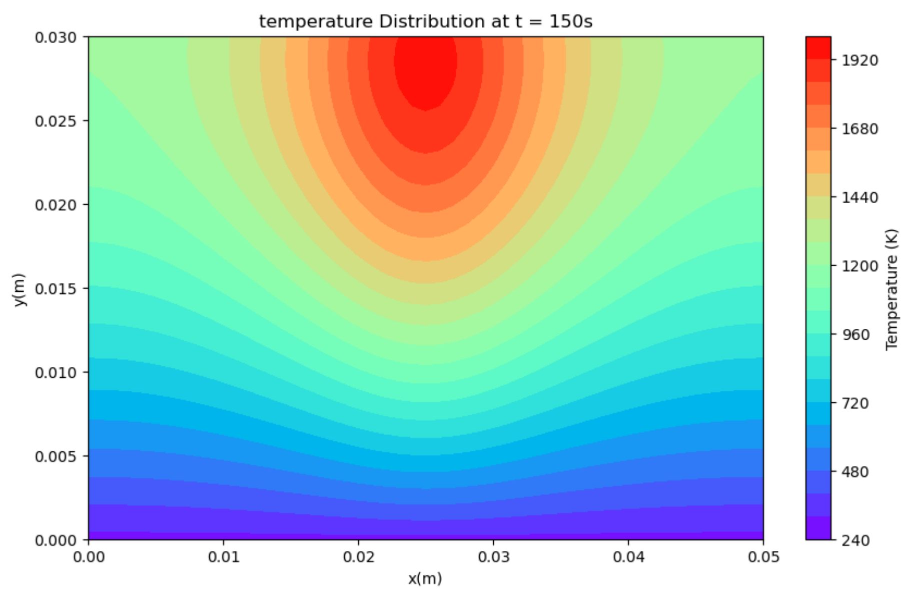

Figure 5. Temperature contour plot at t=150s

Part 5.

The time step chosen in the program is 0.01. By the formula:

$$\Delta t_{max} = \frac{h^2}{4 * \alpha} \tag{22}$$

Where h is the grid spacing, and alpha is provided, the calculated maximum time step is around 0.017, so 0.01 is being chosen.

Part 1 show the table of Maximum temperature at top surface at different time at x=0.5*Lx. From the table, it is obvious that the maximum temperature at top surface increase over time, and the increament rate decrease over time which indicates the system finally approach a stable condition and system converges.

Part 2 shows the Plot of temperature vs time at x=0.5Lx at top surface with different depth. In the graph, top surface has the overall higher temperature, while 50% depth has the lowest oveall temperature. The temperature curves for all depths show a positive temperature vs time relation with decreasing rate, which match and support the observation from part 1.

Part 3 shows the plot of Temperature vs position at the final time along x=0.5Lx, which shows a increaing temperature curve with decreasing rate and has smallest temperature value at bottom (y=0) and maximm temperature value at top (y=0.03), which matches with the observation from part 1 and 2.

Part 4 shows the contour plot of temperature at different time. It can be seen that at t=10s, the heat hasn't transfer very far from the heat source and the maximum temperature is around 1200K, while at t=40s and t=150s, the heat transfer further from the heat source and shows greater maximum temperature. The maximum temperature at t=40s and t=150s are close which match with the previous observation.

Question 3:

Question 3 (20 points):

Write code to model the laser beam traveling with constant velocity. The laser begins at $x = x_{start}$ and ends at $x = x_{end}$, which marks the end of the simulation. Use the parameters listed in Table 3.

Parameter	Description	Value	Units
P	Laser power	800	Watts
v_L	Laser velocity	0.01	m/s
x_{start}	Laser start location	$0.1L_x$	m
x_{end}	Laser end location	$0.9L_x$	m
N_x	$N_x + 1$ grid points in the x direction	50	None
N_y	$N_y + 1$ grid points in the y direction	50	None
t_{end}	End time	Time required to reach x_{end}	sec

Table 3: Simulation parameters for Question 3.

Be sure to use a time step that accurately resolves the laser motion and gives stable, converged results. Defend your choice of time step with calculations and/or relevant simulation results.

Submit the following:

1. (8 pts) Plots of temperature vs time at $x = 0.5L_x$ at the top surface ($y = L_y$), 20% depth ($y = 0.8L_y$), and 50% depth ($y = 0.5L_y$). You may plot all three curves on the same figure if you wish.
2. (8 pts) Contour plots of temperature distribution in the domain at $t = 0.2t_{end}, t = 0.5t_{end}$ and $t = t_{end}$ seconds (3 total plots).
3. (4 pts) Write a few sentences summarizing the key aspects of your simulation results and whether they make physical sense.

Part 1:

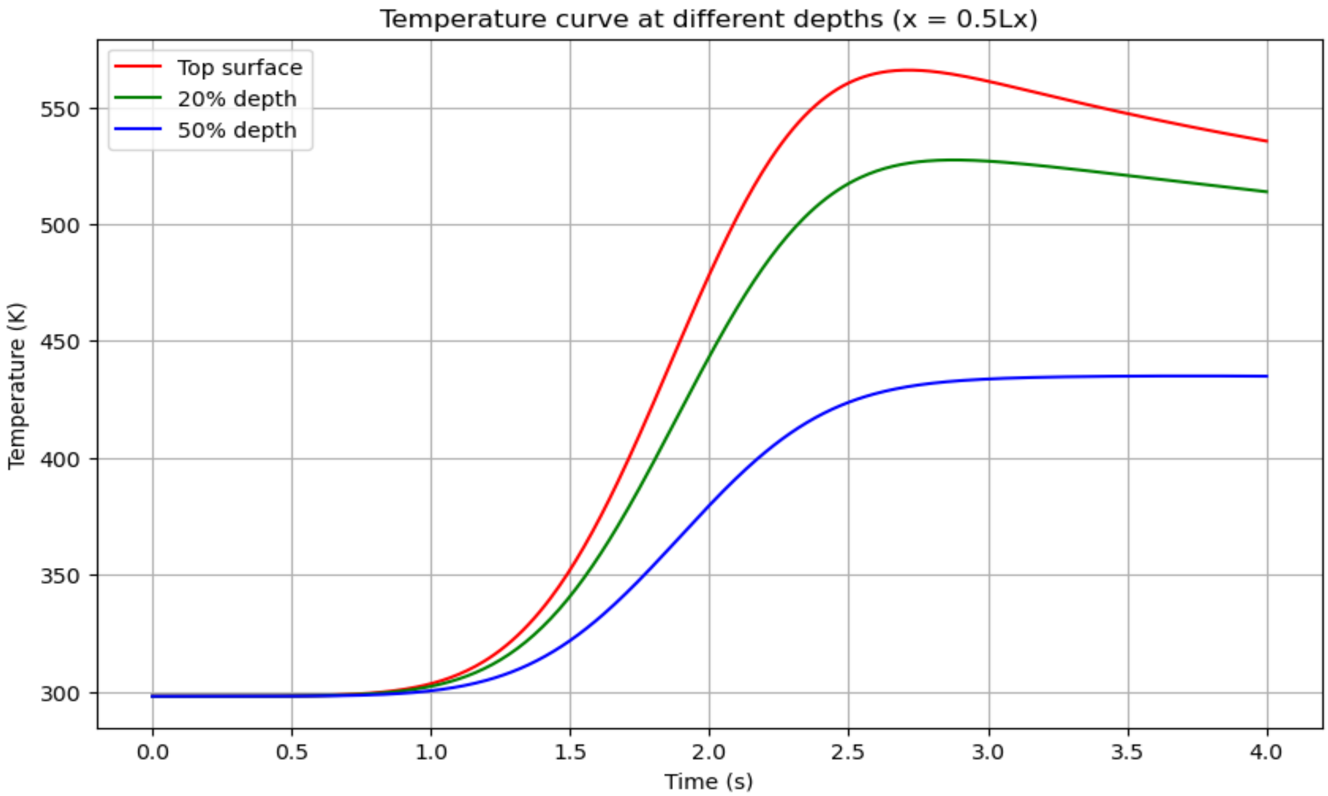

Figure 5. Plot of temperature vs time with moving laser at top with different depth

Part 2:

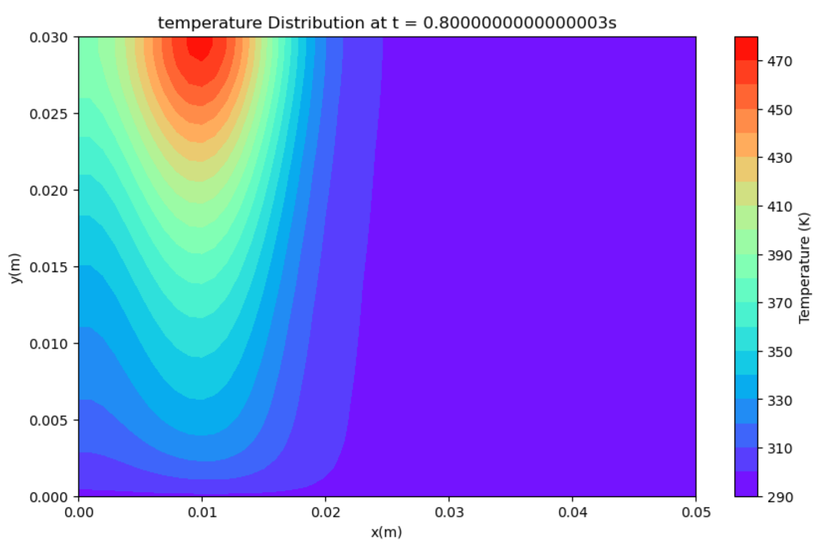

Figure 6. Temperature contour plot at t=0.2*tend

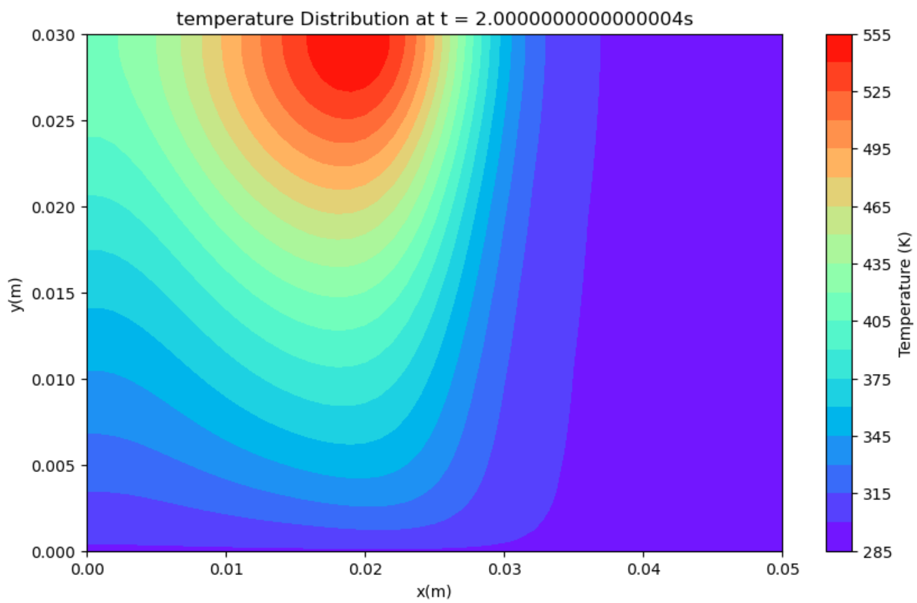

Figure 7. Temperature contour plot at t=0.5*tend

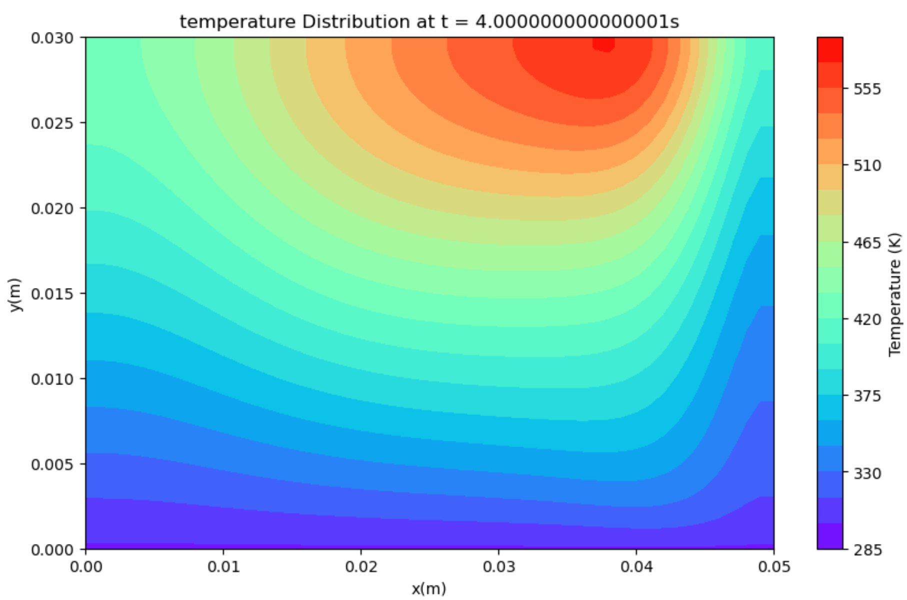

Figure 8. Temperature contour plot at t=tend

Part 3:

Part 1 shows the graph of temperature vs time with moving laser at top with different depth. In the graph, the temperature curves show overall increasing trend with time, with a change in increasing rate at around 2.6s and the curve turn into decreasing rate after 2.6s. From the graph, it can be see that the temperature start to increasing drastically after 1s.

Part 2 shows the contour temperature plots at different time with moving laser. It can be seen that at t=0.8s, the heat hasn't transfer very far from the heat source most heat concentrated left side, while when t=2s and t=4s, the heat has transferred further and the most heat source is concentrated at around middle

Question 4:

Question 4 (20 points):

For a constant velocity straight-line laser scan, investigate the relationship between maximum temperature in the domain versus laser power P and scanning speed v_L for a laser that scans from $x = 0.1L_x$ to $x = 0.9L_x$. The simulation can conclude once the laser reaches x_{end}. Use $N_x = N_y = 50$ For the spatial discretization.

Set up simulations for the following powers and speeds:

$P = 200, 400, 600, 800, 1000, 1200$ W

$v_L = 0.01, 0.02, 0.04, 0.08$ m/s.

(10 pts) For each simulation, record the maximum temperature achieved in the domain over the duration of the simulation. Then, plot this maximum temperature on the vertical axis vs power on the horizontal axis for each of the velocities (4 total curves, 24 total data points, all on one total plot).

(10 pts) Write a few sentences summarizing the key aspects of your simulation results and whether they make physical sense.

Part 1:

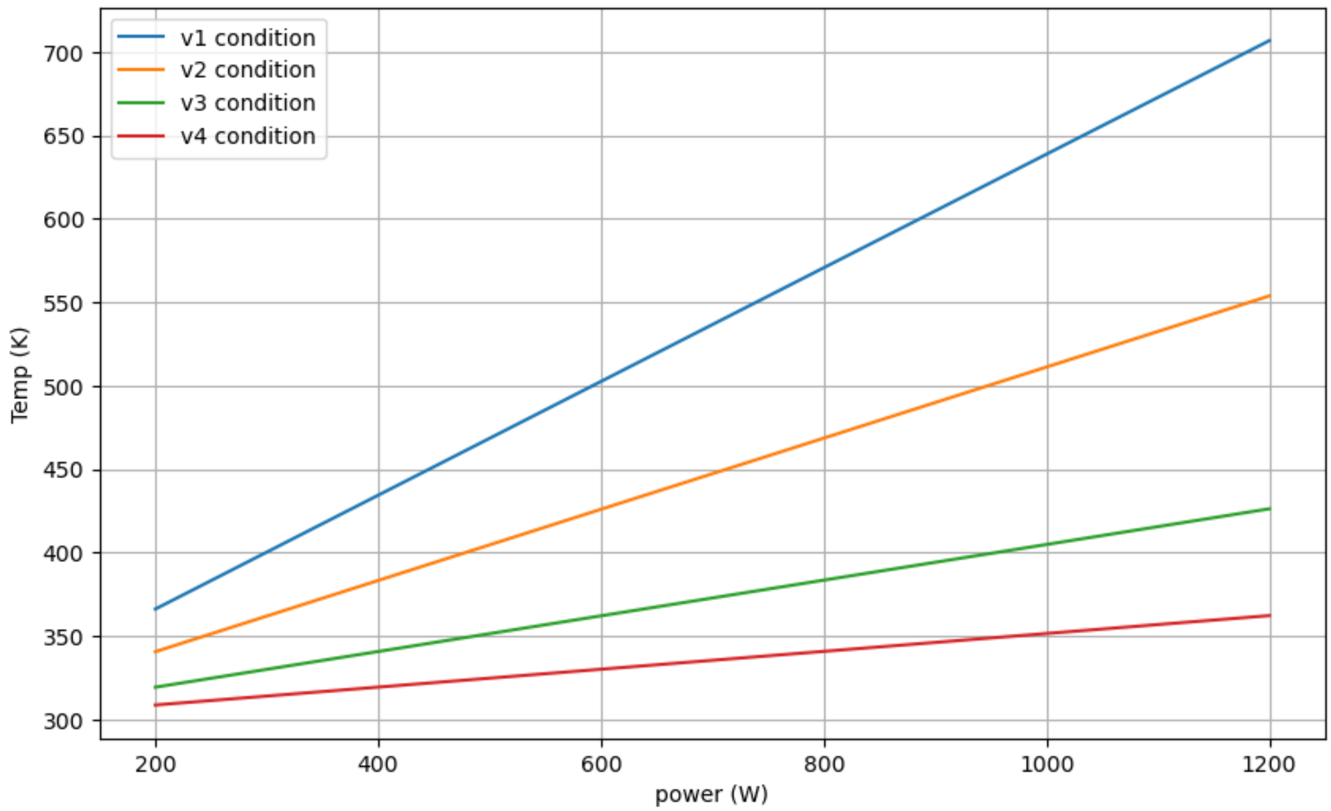

Figure 9. plot of power vs. temperature at different laser velocity

Part 2:

The graph shows that maximum temperature increases linearly with laser power for all scanning velocities. From the graph, it can be seen that lower velocity results in higher overall temperature, which makes sense since lower scanning velocity cause laser heat source stay on the surface for longer time, which result in more heat accumulation that result in higher temperature.